

Effect of Cellulose and Polypropylene on Hydrolysis of Polyethylene Terephthalate for Chemical Recycling

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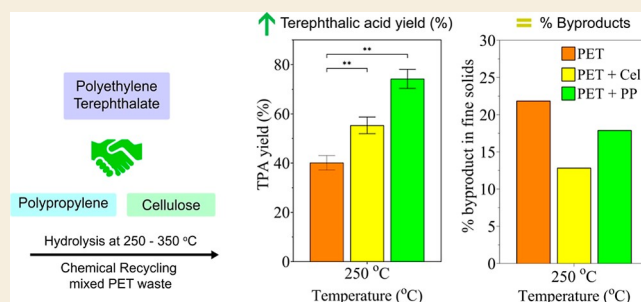
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ABSTRACT: We examined the hydrolysis of polyethylene terephthalate (PET) with added polypropylene or cellulose and measured the yield of the terephthalic acid (TPA) monomer recovered. The TPA yield from hydrolysis at 250 °C for 30 min nearly doubled from 40 to 75% with the addition of polypropylene (PP). It increased to 55% with the addition of cellulose. There were no statistically significant increases in TPA yield from hydrolysis with the added plastic or biomass at 300 or 350 °C. The solid material recovered from the hydrolytic depolymerization, after first recovering water- and dichloromethane-soluble compounds, was largely TPA, and the amounts of the other reaction products present with it were largely the same irrespective of the presence or absence of PP or cellulose in the reactor. The TPA yield was affected strongly by the reaction time, reaction temperature, and PET type (fiber-reinforced pellet vs chips from a water bottle). The addition of PP or cellulose to the reactor reduces the influence of reaction time on TPA yield from PET hydrolysis.

KEYWORDS: chemical recycling, plastics, waste valorization, circular economy, hydrolysis, hydrothermal



INTRODUCTION

Polyethylene terephthalate (PET) is a plastic material predominantly used for fibers, fabrics, and single-use food and beverage packaging. The majority of PET waste (~75%) is landfilled. Only ~12% is recycled and 3% is burned as fuel for energy recovery.¹ The need for separation and sorting of waste by type is a limitation to the growth of recycling.² Further, conventional mechanical recycling of plastics downgrades PET properties and reduces its value. Recycled PET is not further recycled but discarded in landfills.³

To achieve a circular economy for PET, chemical recycling needs to be employed. Early approaches to recover monomers from PET, developed in the 1960s and 70s, typically involved glycolysis, which gave bis (2-hydroxyethyl) terephthalate (BHET) as the main product.^{4,5} The process has been commercialized, primarily in Europe.^{6,7} However, the cost of ethylene glycol and corresponding carbon footprint are negative factors. Methanolysis produces dimethyl terephthalate (DMT) from depolymerized PET.⁸ This process found commercial applications for recycling scrap PET produced during product manufacturing.⁹ However, terephthalic acid (TPA) is preferred over DMT as the product from chemical recycling of PET.¹⁰ Using water as the reaction medium for PET depolymerization produces TPA as the main product, and hydrolysis in water is a potentially greener alternative to glycolysis and methanolysis.³

Hydrolysis of PET in liquid water above its normal boiling point gives TPA and ethylene glycol as the main products.³

Work in this field dates back at least to patents from the 1980s.^{11–13} The patents highlight the possibility of recovering both TPA and ethylene glycol from hydrolysis of PET wastes in aqueous reaction medium at 200–300 °C. Mancini and Zanin¹⁴ conducted optimization experiments to obtain 99% TPA yield from hydrolysis of PET at 205 °C for 6 h. The size of the PET particle did not influence the quality of the product. Zhang¹⁵ reported that pre-treatments such as milling and photo-aging do not influence the reactivity of PET during hydrolysis. The literature also contains several kinetics studies for hydrolysis of PET waste.^{16–21} These kinetic experiments were conducted either at lower temperatures (e.g., 180 °C) and long times (>4 h)^{18–21} or at higher temperatures (e.g., 260 °C) and short times (≤30 min).^{16,17} The shorter times are desirable for commercial processes for hydrolysis of PET at scale. The processing temperatures can be reduced by adding either alkaline or acid catalysts.^{22,23}

Most previous works on chemical recycling of PET via hydrolysis were conducted with PET waste alone. As discussed above, however, the need for separation and sorting of waste materials has curbed mechanical recycling and has the

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potential to limit chemical recycling as well if PET alone is needed as the feedstock. This realization has motivated the present study into the hydrolytic depolymerization of PET when present with another plastic (polypropylene) or with a biomass component (i.e., cellulose). We selected polypropylene (PP) as it is typically used for the caps on single-use PET bottles. We selected cellulose because polysaccharides are major components of MSW (e.g., waste paper, food waste, and yard trimmings).¹ The only prior work on PET hydrolysis in a mixture of this type of which we are aware is from Palme et al.,²² who decomposed cotton and PET together from mixed fabrics using alkaline hydrolysis with a phase-transfer catalyst. That work focused on optimizing the process and catalyst and reported 99% TPA yield at 90 °C and 40 min with 10% NaOH and 52 mmol of phase-transfer catalyst/kg solution. The authors did not report any experiments with PET or cotton alone, so there is no information about whether these two components interacted during the hydrothermal treatment. Previous works on hydrothermal treatment of macromolecules showed that interactions can occur among plastics and between plastics and biomass components, which promote depolymerization and increase oil yield.²⁴ We hypothesize that such interactions could also occur during PET hydrolysis and influence monomer yields obtained from PET when processing it as one component in a mixture of waste materials. Thus, we focus herein on the TPA yields obtained while co-processing PET with another plastic and with a biomass component. We identify the process variables that most influence TPA yield when PET is processed alone and in mixtures. We further use simpler model compounds to test hypotheses regarding interactions between PET and PP or cellulose.

MATERIALS AND METHODS

Materials

PET cylindrical pellets (3.64 ± 0.26 mm $\times$ 2.37 ± 0.24 mm) and PET single-use water bottles were obtained from Sigma-Aldrich and Walmart, respectively. The cylindrical pellets were reinforced with glass fibers. Thermogravimetric analysis indicates 39.8 ± 0.8 wt % of the PET pellets is from glass fibers (see Figure S1), as the PET itself would leave no residual mass above 800 °C.²⁵ The cap and wrapper of the PET bottles were removed, and the bottle was cleaned. The central part of the bottle was then cut into square chips (1.2 cm $\times$ 1.2 cm), which were then used in the experiments. Polypropylene pellets (Sigma-Aldrich), cellulose powder (microcrystalline, Alfa Aesar), glucose powder (D(+), dot scientific), squalane (Alfa Aesar), glass beads (3 mm, Sigma-Aldrich), and tetralin (TCI) were used as received.

Deuterated dimethyl sulfoxide (DMSO) with 1 wt % TMS was purchased from Cambridge Isotope. Reagent grade dichloromethane and DMSO were from VWR International, and deionized (DI) water was prepared in-house with $\sigma = 18 \pm 2$ μ S. Helium cylinders were acquired from Penn State general stores. 10 mL (internal volume) stainless-steel mini-batch reactors were made from nominal 3/4 in. Swagelok port connectors and caps.

Hydrolysis Procedure

The reactors were loaded with the amount of feedstock that gave the desired ratio with water. DI water was added in different amounts depending on the desired hydrolysis temperature (7.59 mL for 250 °C, 6.15 mL for 300 °C, and 4.96 mL for 350 °C). The water loadings were sufficient to maintain a liquid phase in the reactor at the reaction temperature while also providing a safety factor and avoiding over-pressurization of the reactor. PET and either PP or cellulose were added in equal amounts for the mixture experiments. The loaded reactors were placed in a preheated isothermal Techné fluidized sand bath at setpoint temperatures of 250, 300, or 350 °C for 30, 45, or 60

min. Many experiments were performed in at least triplicate, and we report the mean value and the standard deviation as an estimate of the run-to-run variability.

After the desired holding time had elapsed, the reactors were removed from the sand bath and quenched in water at room temperature for 10 min. The reactors were further equilibrated at room temperature for 24 h prior to opening. The reactor contents were recovered first by using 10 mL of dichloromethane (DCM) and 10 mL of deionized water in smaller successive aliquots. We refer to the DCM-soluble material as oil and the water-soluble material as aqueous-phase products. All of the liquid aliquots were withdrawn from the reactor by pipettes and passed through a 0.45 μ m cellulose fiber filter from GE Research to collect the finer solid material suspended in these solutions. Larger solid particles that were not suspended in the solutions remained in the reactor. After extraction with DCM and water, the filters and reactors were dried in an oven at 60 °C overnight and then weighed to determine the mass of solids in each.

TPA Analysis

After drying, the larger solid particles remaining in the reactor were dissolved in 10 mL of DMSO, which then passed through a filter to remove any unreacted PET and other insolubles. The amount of TPA in a given filtered DMSO solution was determined using a Shimadzu high-performance liquid chromatography (HPLC) system equipped with a Waters Symmetry C18 column (5 μ m particle, 150 mm $\times$ 4.6 mm). A SPD-M20A PDA detector was used at 254 nm. The column and the detector were maintained at 40 °C. The mobile phase was a mixture of 1 vol % H₃PO₄ aqueous solution (0.3 mL/min) and HPLC grade acetonitrile (0.1 mL/min). The injection volume was adapted between 1 and 10 μ L based on the TPA concentration in a given sample. Calibration curves were generated to quantify the TPA product based on analysis of TPA standard solutions with known concentrations. This HPLC method is adapted from the literature.²⁶

A portion (10 mg) of the fine solids that had been trapped on the filter was dissolved in 0.6 mL of deuterated DMSO with 1 wt % TMS and placed in an NMR tube. Because the volume of the solvent used here is much less than that needed for the solids remaining in the reactor, we used the more costly deuterated DMSO and opted for NMR analysis rather than HPLC. The NMR tube is weighed to calculate the mass of solids and solvent added gravimetrically. A Bruker NMR DPX400 analyzer set at 400 MHz was used to obtain the ¹H NMR spectrum with a protocol developed in house. TMS served as the reference standard ($\delta = 0$ ppm) and internal standard for quantifying the amount of TPA. The following instrument settings were adopted: pulse length (90°) = 7.8 μ s, delay = 32 s, number of scans = 72, spectral width = 4800 Hz. Quantitative results were obtained by analyzing standard solutions with 1, 5, and 10 mg of TPA dissolved in the same solvent. The peak area for the aromatic protons in TPA was compared with the TMS peak area to quantify the TPA concentration, which was then used to calculate the mass of TPA in the fine solids. We calculate the percentage of byproducts in the fine solids as the sum of the areas of the peaks from byproducts divided by the total area (byproducts plus TPA).

The total amount of TPA recovered from a hydrolysis experiment is the sum of the TPA determined by HPLC analysis of the solid residue that remained in the reactor and the TPA determined by NMR analysis of the fine solids trapped in the syringe filter. Tables S1 and S2 show the percentage of the total TPA that was recovered in the fine solids for each run. The TPA yield is calculated as the percentage of the maximum yield possible given the stoichiometry of the hydrolytic depolymerization reaction, the amount of PET loaded into the reactor, and assuming complete depolymerization (see eq 1).

$$\text{TPA yield(\%)} = \frac{\text{Total TPA measured}}{\text{Theoretical maximum TPA in sample}} \times 100 \quad (1)$$

To estimate the fraction of the protons in the DMSO-solubles that are not in TPA, we sum the areas for the peaks corresponding to

protons in the byproducts and divide that quantity by the sum of the peak areas for all of the products (eq 2).

$$\text{Byproduct}(\%) = \frac{\sum \text{Area of byproduct peaks}}{\sum \text{Area of all product peaks}} \times 100 \quad (2)$$

This byproduct level, though not identical to the wt % purity of TPA, is useful for assessing the presence of compounds other than TPA in the DMSO-solubles.

Thermogravimetric Analysis

Thermogravimetric analysis of powdered PET pellets was conducted on a NETZSCH STA 449 F3 Jupiter thermal analyzer in an inert nitrogen environment. The temperature increased from 25 °C to 1000 °C at 20 °C/min. The residual mass (%) at 1000 °C is taken to be the wt % of the glass fiber initially present in the PET pellets. The TGA analysis was performed in triplicate. We use the mean value as the best estimate of the true value and the deviation as an estimate of the uncertainty (see Figure S1).

pH Values of Slurry after Hydrolysis

We determined the pH (at ambient conditions) of the slurry after hydrolysis of cellulose alone, glucose alone, and PET alone at 250 °C for 30 min. A Thermo Scientific Orion Versa Star pH meter and Ross Ultra pH/ATC triode electrode were used for measuring the pH of solutions. Calibration and accuracy were maintained by using buffer solutions (pH 4, 7, and 10). The contents of the reactors were removed and diluted fivefold with water so there would be sufficient samples for the pH meter. The pH of this solution is measured and used to calculate the pH of the slurry.

RESULTS AND DISCUSSION

Effect of Added Plastic or Biomass on PET Hydrolysis

Figure 1 shows the presence of cellulose or PP during hydrolysis of PET pellets can change the yield of TPA. For

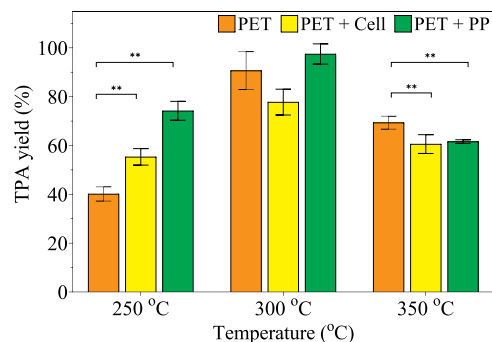


Figure 1. TPA yields from hydrolysis of PET pellets alone and in equi-mass mixtures with PP and with cellulose at different temperatures for 30 min. Reactor loadings (g feedstock/g water) were 4.9, 6.0, and 7.5% at 250, 300, and 350 °C, respectively. ** indicates a statistically significant difference ($p < 0.025$) between the runs with PET alone and with additives. Table S3 gives the yields of solid and oil from these experiments.

hydrolysis at 250 °C, both materials have positive interactions that increase the TPA yield. At 250 °C, PET does not fully depolymerize to form monomers after 30 min. The TPA yield is only 40% when PET is processed alone, but it increases to 75% when PP is added. A separate set of experiments was run with PET chips from water bottles. Hydrolysis (250 °C, 15 min) of PET chips alone, with added PP, or with added cellulose gave TPA yields of 34.8 ± 0.6 , 66.0 ± 1.6 , and $69.4 \pm 2.4\%$. Thus, both PET sources exhibited synergistic depolymerization when processed as part of an equi-mass mixture with PP or cellulose. These materials accelerate the hydrolysis

process and increase the TPA yield. We will return to these synergies and potential mechanisms for them in a subsequent section of this article. Also note that TPA has some limited solubility (0.0015 g/100 mL) in water at room temperature. If the roughly 18 mL of water that was used in each experiment (at 250 °C) was saturated with TPA, it would add only another 0.16 wt % to the TPA yields reported herein.

The TPA yields in Figure 1 from hydrolysis of PET at 300 °C and 30 min range from 80 to 95%. These high values indicate nearly complete depolymerization of PET to its monomers and little opportunity for added PP or cellulose to have any effect in accelerating the depolymerization. Hence, the additives have no statistically significant effect at this temperature.

Figure 1 also indicates statistically significant decreases in TPA yield from hydrolysis of mixtures at 350 °C for 30 min when compared to hydrolysis of PET alone at the same conditions. Because PET depolymerizes nearly completely after 30 min at 300 °C, higher temperatures may cause the TPA product to decompose. The added PP or cellulose appears to modestly promote the decomposition process as well. Previous works showed hydrothermal decarboxylation of TPA to benzoic acid at 350 °C.²⁶ This decomposition would result in lower TPA yields in the present experiments, whether PET is processed alone or in a mixture. Benzoic acid is soluble in DCM, so we expect this decomposition product to partition into the oil phase and not the TPA-containing solids recovered with DMSO.²⁷

Prior works have reported similar TPA yields from neutral hydrolysis at similar temperatures. Rosen¹² processed post-consumer PET bottles at 300 °C for 6 h and obtained a 93% TPA yield. Mancini and Zanin¹⁴ did neutral hydrolysis of post-consumer PET flakes obtained from a recycler at 205 °C for 6 h and obtained 99% PET conversion. Using shorter reaction times gave lower TPA yields, which is consistent with other works¹⁵ that reported TPA yields ~11% from hydrolysis at 200 °C.

The literature noted above shows that longer times at a given temperature can give higher TPA yields from hydrolysis. Depolymerization at 250 °C with times longer than the 30 min used herein would likely have given higher TPA yields, as would hydrolysis at temperatures lower than those investigated herein (e.g., 200 °C) and long times. Determining the effects of PP, cellulose, and other additives on PET hydrolysis at 250 °C, or even 200 °C, and longer reaction times could lead to identifying a lower operating temperature that is nevertheless effective for essentially complete depolymerization of PET. Evidence to support this hypothesis comes from the work of Palme et al.,²² who evaluated the TPA yield from alkaline hydrolysis of PET-cotton mixtures. They obtained a 95% TPA yield at 90 °C, which is much lower than the 150–200 °C temperature range where alkaline hydrolysis typically occurs for PET alone.^{23,28}

Quality of TPA Produced

The method used herein to recover the reaction products separates the TPA from many other coproducts by first using DCM and water to dissolve products from the reaction. Even so, there are likely other compounds formed that, like TPA, are insoluble in DCM and water but soluble in DMSO. In this section, we use the NMR results to estimate the fraction of the protons in the DMSO-soluble material in the fine solids that are not in TPA. Section 1 of the Supporting Information gives

a sample calculation. The NMR spectra in Figure S2a–c show that processing PET at 250 °C alone or with PP or cellulose produces compounds other than TPA, but the spectra are all similar irrespective of the presence or absence of an additive. For hydrolysis at 300 or 350 °C, the NMR peaks corresponding to byproducts are smaller than they are from hydrolysis at 250 °C (Figure S2d–i).

Figure 2 shows that the byproduct levels are lower for hydrolysis of PET with PP or cellulose at 250 °C than for

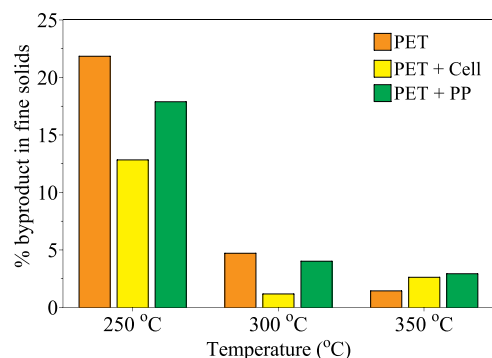


Figure 2. Byproduct % in fine solids from hydrolysis of PET pellets alone and in equi-mass mixtures with PP and with cellulose at different temperatures for 30 min. Reactor loadings (g feedstock/g water) were 4.9, 6.0, and 7.5% at 250, 300, and 350 °C, respectively.

hydrolysis alone. Thus, the additives not only increased the yield of TPA, but they also decreased the byproducts recovered alongside it in the experimental protocol. The byproducts are present at an even lower level from hydrolysis at 300 and 350 °C. At these higher temperatures, PET hydrolysis is essentially complete, so there are fewer intermediate byproducts. The results from hydrolysis at 350 °C indicate that the products from the decomposition of TPA (a hypothesis to explain the TPA yields being lower at 350 °C than at 300 °C) are water- or DCM-soluble and are not concentrated in the DMSO-soluble solid phase. Benzoic acid is one potential decomposition product from TPA that fits this profile. Had benzoic acid been present in the fine solids, it would have been detected by the NMR method employed (see Figure S3). Given the solubility of benzoic acid in DCM and water, we would not expect it to remain with TPA in the water- and DCM-insoluble fractions.

A significant finding in Figure 2 is that the presence of PP or cellulose during PET hydrolysis does not increase the amount of byproducts in the solids containing the TPA. Cosimbescu et al.²⁹ studied the byproducts from hydrolysis (with ethylene glycol or ethanol as co-solvents) of pure PET and mixed waste (PET bottles with caps). Their analyses indicated 97–98% purity (based on proton peak areas from NMR) of the TPA irrespective of the components in the initial mixture. These values align with what we report in Figure 2 for hydrolysis at 300 and 350 °C. This lack of influence on byproducts, if generally true for other plastics and biomass components likely to be in mixed waste streams, would be an important advantage for hydrothermal treatment of such materials as it would remove the need for sorting and separating the different waste types prior to treatment, which is a significant deterrent for mechanical recycling.²

The byproduct levels from hydrolysis at 300 and 350 °C based on proton detection are low enough to be encouraging, but not low enough to use the TPA “as is” to synthesize new PET. A purified terephthalic acid with >99 wt % purity is desired.^{30,31} Thus, some additional purification steps (e.g., crystallization) might be needed to complete this chemical recycling of PET and produce resin quality TPA.

Factors Influencing TPA Yield

Having demonstrated that TPA can be recovered in nearly 100% yield and with just a few % of byproducts from hydrolysis of PET mixed with PP or cellulose, we next identify the process variables that would most strongly influence the TPA yields from hydrolysis of PET alone and in mixtures. We designed hydrolysis experiments, as outlined in Table 1, to test the significance of temperature, time, feedstock loading, and PET type (chips from a bottle, pellet with glass reinforcement) on the TPA yield and % byproducts.

We used a fractional factorial design (2^{4-1}) with the quaternary interaction effect confounded to take the high value for the experiments. The confounded factors are listed in Section 2 of the Supporting Information. We are primarily interested in the effects of the individual factors, and this fractional factorial design serves that purpose. We added two center points to the design (runs 9 and 10) to test the linearity of the factor-response dependence. Because factorial designs assume linear relationships between factors and responses, the absence of curvature is a prerequisite for analysis using the method. It indicates that maxima of the response variables (e.g., TPA yield) do not lie between high and low values of

Table 1. Experimental Design for Determining the Effects of Temperature, Time, PET Type, and Feedstock Loading on TPA Yields from PET Hydrolysis

DOE exp. order	temperature (°C)	time (min)	PET type	feedstock loading (g feedstock/g water) (%)	TPA yield (%)			% byproducts		
					PET alone	PET + PP	PET + cellulose	PET alone	PET + PP	PET + cellulose
1	250	30	pellet	5	19.66	21.09	10.74	40.70	38.29	38.68
2	350	30	pellet	20	52.78	94.05	76.22	4.33	7.36	2.26
3	250	60	pellet	20	51.47	15.54	0.11	9.30	10.34	21.61
4	350	60	pellet	5	29.12	60.32	85.02	6.88	3.85	4.91
5	250	30	bottle	20	0.03	47.73	61.82	49.80	40.20	17.75
6	350	30	bottle	5	55.03	69.83	40.15	3.31	1.67	1.48
7	250	60	bottle	5	85.94	44.82	68.93	3.00	5.67	2.68
8	350	60	bottle	20	85.42	81.42	48.43	6.00	5.32	2.64
9	300	45	pellet	12.5	45.17	92.37	21.03	7.64	32.90	39.79
10	300	45	bottle	12.5	48.79	31.79	79.01	12.41	2.84	1.64

process factors. The p -value for curvature (0.932) is much greater than 0.05 (see ANOVA Table S4), confirming the absence of statistically significant curvature in the designed experiments with PET alone. The same is also the case for the designed hydrolysis experiments for PET with PP and with cellulose (see ANOVA Tables S5 and S6). Figure S4 provides a visualization of the TPA yields (%) from hydrolysis of PET alone across the variable space. We transformed the temperature factor into $e^{-298K/Temp.(K)}$ as we expected T alone to have an exponential, Arrhenius-type effect on the TPA yield. Also, if a certain factor has a p -value greater than 0.9, it is removed from the ANOVA analysis and tables in the Supporting Information.

Table 1 gives the experimental results for the designed experiments, which were then used with different statistical tools to assess the importance of different process variables. Normal plots, such as Figure 3 for hydrolysis of PET alone,

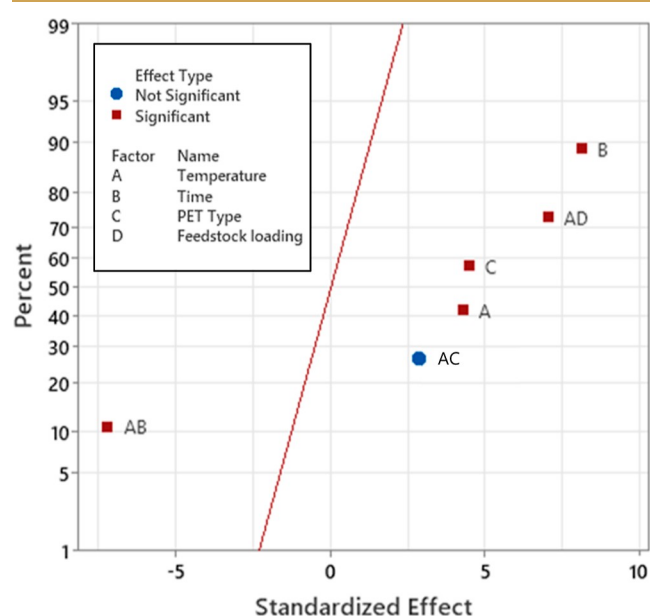


Figure 3. Normal plot for standardized effects of process variables on TPA yield from hydrolysis of PET alone ($\alpha = 0.05$ for significance).

show standardized effects relative to a distribution fit line when all effects are zero. The value for the standardized effect is equivalent to the t -statistic for each coefficient in the ANOVA table (see Table S4). A positive standardized effect of a factor indicates increasing monotonic dependence with the response (TPA yield) and vice versa. A t -test is conducted with $\alpha = 0.05$ to ascertain the significance of a factor. Factors that are not statistically significant do not appear on the plot. More information about the plot is available elsewhere.³²

PET type, temperature, and time are significant factors in Figure 3 (represented as red squares) that positively correlate with TPA yield. The positive effect of temperature is consistent with the results in Figure 1 showing an increase in TPA yield as the hydrolysis temperature increases from 250 °C. The standardized effect of time on the yield exceeds that of the temperature. This relative ranking is consistent with literature¹⁴ observations of high TPA yields at lower process temperatures and longer process times. The PET type is set up in the factorial design such that the bottle is assigned a high value and the pellet is assigned a low value. Thus, transitioning from a pellet to a bottle under similar conditions would increase TPA yield. This influence is reasonable as the PET pellets used herein contain glass fiber reinforcers that would likely interfere with the depolymerization of the pellet.^{33,34} Separate hydrolysis experiments (250 °C, 30 min) with chips from the PET bottles gave a high TPA yield of $77.3 \pm 3.0\%$. This TPA yield exceeding the 40% yield from hydrolysis of the pellets at the same conditions is consistent with the outcome from the designed experiments. The extent of PET conversion via hydrolysis has been previously linked to its type or form. Ügdüler et al.³⁵ found the TPA yield from alkaline hydrolysis of different types of PET (pure pellets, bottles, and films) to be in the order bottles < films < pure pellets. The PET pellets in that study gave the highest yields, but they were pure resin and did not contain glass fibers. This difference in the pellets likely accounts for their lower reactivity in the present work.

Feedstock loading does not appear on Figure 3 as it is not a significant factor influencing TPA yield during its hydrolysis alone, likely because water was always present in large excess. For a repeat unit of PET to hydrolyze to TPA, it needs a molecule of water which amounts to the stoichiometric feedstock loading being 1250% g PET/g water. Because the highest feedstock loading (g PET/g water) used in the experiments is just 20%, water is always present in large excess, and the feedstock loading is not an influential factor.

Several interaction factors are significant in influencing the TPA yield from hydrolysis of PET alone and with PP or cellulose (see Figures 3 and S5), but these interaction effects are confounded with each other. It is not possible to know which of these interactions is causing the term to be significant.

For hydrolysis of PET with PP and with cellulose, temperature remains a significant factor (see Figure S5), but time does not. The promotion of degradation of PET by PP and cellulose seems to have overshadowed the effect of time. For hydrolysis of PET with cellulose, the PET type is a significant factor influencing the TPA yield. PET bottles degrade faster with added cellulose than PET pellets do. This outcome is consistent with the observation made for hydrolysis of PET alone. However, the PET type is not significant in influencing the TPA yield for hydrolysis of PET with PP. As

Table 2. Optimal Process Conditions within Low/High Limits for Simultaneously Maximizing TPA Yield and Minimizing % Byproducts as per ANOVA Analysis from Experimental Design

process condition	PET alone		PET + PP		PET + cellulose	
	bottle	pellet	bottle	pellet	bottle	pellet
temperature (°C)	350	350	350	350	350	350
time (min)	60	60	30	30	60	60
feedstock loading (% g PET/g water)	5	5	20	20	5	5
ANOVA predicted TPA yield (%)	84.5	80.0	86.7	103.0	77.1	80.6
ANOVA predicted byproducts (%)	2.72	3.91	5.27	5.27	−3.44	12.77

discussed in Figure 1, PP increases the TPA yield more than cellulose does during PET hydrolysis. This greater effect of PP could result in both PET pellets and PET bottles giving similar TPA yields when hydrolyzing PET-PP mixtures.

Examining results for the percentage of byproduct protons from PET hydrolysis shows that for hydrolysis of PET alone or with PP or with cellulose, time and temperature are significant factors (see Figure S6). These factors have negative effects, which means that increasing time or temperature leads to a decrease in byproducts. This observation from these designed experiments is consistent with the results in Figure 2. PET type is not a significant factor for the presence of byproducts.

Table 2 shows the ANOVA results of individual factors for maximizing the TPA yield and minimizing the percentage of byproducts. Equal weight is given to both responses to calculate the composite desirability of a given condition while performing the optimization. When PP is added to PET, the reaction time for maximizing the TPA yield is reduced from 60 to 30 min. This reduction in time is further evidence of the presence of interactions that increase the yield of TPA from PET during hydrolysis. Because cellulose provides less synergy than PP, the process factors for the optimal conditions remain the same as for the case of PET alone. All of the optimal points are in orthogonal regions of the process factors, that is, in the extremes, which is consistent with results above showing the TPA yield and percentage byproducts are inversely related. This work with designed experiments has given insights into the relative influence of the different process variables for PET hydrolysis, and the results affirm the presence of synergistic interactions between PET and PP that increase the TPA yield and decrease byproducts during hydrolysis.

Probing Interactions during PET Hydrolysis

Earlier, we showed that the presence of PP and cellulose accelerates the decomposition of PET to monomer. In this section, we generate and assess hypotheses regarding the origin of the observed synergy. Figure 4 gives the results from hydrolysis experiments conducted to test the hypotheses. All experiments were conducted at 250 °C for 30 min using PET pellets and an equal mass of the second component.

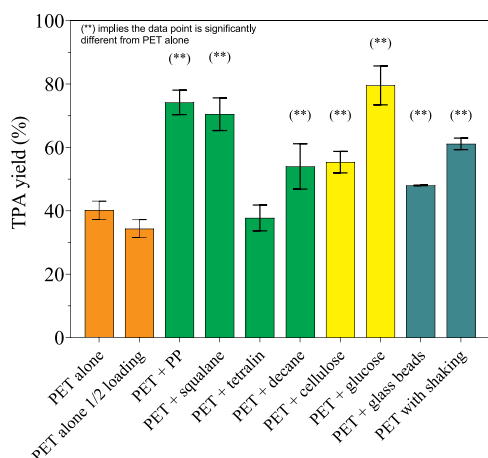


Figure 4. TPA yields from hydrolysis (250 °C, 30 min) of PET alone and with additives. The ** indicate a significant difference when compared with hydrolysis of PET alone ($\alpha = 0.05$ for significance). The orange bar is the base case experiment, while the light green, yellow, and dark green bars correspond to experiments testing hypotheses about H-transfer, pH, and transport, respectively.

Experiments were replicated three times and we report the mean and standard deviation.

Noting that the materials that accelerated TPA production are more hydrogen rich (PP is 14.2 wt % H, cellulose is 6.2 wt % H) than PET (3.8 wt % H) suggesting that perhaps these additional hydrogen atoms are somehow playing a role. One possible role is that of hydrogen transfer where the more hydrogen-rich molecules serve as donors to the PET as it decomposes. To test whether classical hydrogen transfer might be involved, we performed the hydrolysis of PET with added 1,2,3,4-tetrahydronaphthalene (tetralin). The structure of tetralin is given in Figure S9a. This molecule is a well-known H-donor, due to its containing four weakly bound benzylic protons. Figure 4 shows the added tetralin had no effect on the TPA yield, indicating that classical free-radical H-transfer (e.g., abstraction of H from the donor molecule) does not seem to be playing a role. This outcome is reasonable, as we expect PET decomposition in the present experiments to follow a molecular hydrolysis mechanism rather than a free-radical mechanism.

Having ruled out hydrogen donation as the cause of synergy, we next tested whether the synergy was connected to the chemical structure of PP. We conducted the hydrolysis of PET with added 2,6,10,15,19,23-hexamethyltetracosane (squalane), as a small-molecule (C_{30}) mimic of PP (see Figure S9b for molecular structures). Figure 4 shows that the addition of squalane enhanced the TPA yield ($p < 0.05^{**}$) from PET hydrolysis to essentially the same extent as did PP. This outcome suggests there is something about the chemical structure or composition of the methyl-branched alkanes that leads to the synergy. We further tested this with the addition of decane (C_{10}) to PET. There is a smaller yet significant increase in TPA yield due to the addition of the n-alkane.

Noting that added cellulose promoted TPA formation and that the hydrothermal decomposition of cellulose produces compounds that are acidic,^{36,37} its decomposition could be reducing the pH of the hydrothermal medium and thereby accelerating the hydrolysis of PET. To determine whether cellulose-decomposition products might be influencing the TPA yield, we performed hydrolysis of PET with added glucose, the monomer in cellulose. Its hydrothermal decomposition should give the same reaction products as those from cellulose. Figure 4 shows the added glucose enhanced TPA production ($p < 0.05^{**}$) even more than added cellulose did. This outcome is consistent with pH reduction leading to the synergy observed. The pH of the slurry from hydrolysis of PET–cellulose and PET–glucose mixture at 250 °C, 30 min are 2.41 and 2.10, respectively. The more acidic pH resultant from hydrolysis of glucose in PET–glucose mixture is consistent with the greater synergy observed from hydrolysis of PET mixtures with glucose when compared to cellulose. Note however that slurries from hydrolysis of both these carbohydrates alone have a more acidic pH than from hydrolysis of PET alone at 250 °C (3.18).

Finally, we note that hydrothermal treatment of PET with added cellulose or PP would likely involve the presence of solids in the reactor (e.g., char from cellulose and unreacted PP) that would not be present during hydrolysis of PET alone. Perhaps the additional solids could have promoted mixing and heat or mass transport within the reactor, thereby accelerating hydrolysis of PET. To test whether the presence of additional solid particles would influence the TPA yield, we performed the hydrolysis experiment with added glass beads. Figure 4

shows the glass beads gave a modest, but statistically significant, increase in TPA yield. To test more directly whether mass transfer might be limiting the formation of TPA, we conducted a hydrolysis experiment wherein the reactor was attached to a wrist-action shaker and shaken continuously throughout the 30 min experiment. This run gave a higher TPA yield and affirmed that additional mixing enhances the hydrolysis reaction.

To summarize, we believe there are chemical interactions involving the branched alkane structure of PP and squalane and chemical interactions involving glucose and/or its decomposition products that enhance hydrolysis of PET and formation of TPA (see Figure S10). The precise nature of these interactions is unknown at present and is the topic of ongoing studies. We also showed that the reactions were done in a transport-limited regime and better mixing enhanced TPA yields.

CONCLUSIONS

- 1 TPA can be recovered in yields of about 80% or greater and with low byproduct levels from hydrolysis of PET even when it is mixed with another plastic (polypropylene) or with biomass (cellulose). Additionally, the presence of PP or cellulose in the reactor increased the TPA yield from hydrolysis of PET at 250 °C, indicating they had a synergistic effect. Finally, co-processing PET with PP or cellulose did not influence the levels of impurities in the TPA product. These positive outcomes suggest that this hydrothermal approach may facilitate chemical recycling of PET in mixed wastes by reducing the need for separating and sorting waste materials.
- 2 Temperature, PET type (pellets or chips), and reaction time are the factors that most influence the yield of TPA from hydrolysis of PET. Pellets degraded at a slower rate than chips did from PET bottles, likely due to the presence of glass fiber reinforcers in the pellets.
- 3 The addition of PP or cellulose to PET reduces the influence of reaction time on the TPA yield from this hydrothermal chemical recycling process. The addition of PP also reduces the effect of PET type on TPA yield.
- 4 The precise causes for synergy during PET hydrolysis are not yet clear, but methyl-branched alkanes and glucose and/or its decomposition products seem to interact chemically. Physical causes may also have been at play in the experiments as increased mixing led to higher TPA yield than hydrolysis did in a stationary batch mini-reactor.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsengineeringau.2c00024>.

Percentage TPA yields from fine solids on filter, oil, and solid yields, ANOVA tables for all designed experiments, TGA results for PET pellets, NMR spectra, cube plot with experimental conditions and TPA yield for designed experiments, normal plots for effect on percentage byproducts for all designed experiments, molecular structures, and summary of PET hydrolysis reaction (PDF)

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Notes

The authors declare no competing financial interest.

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